

Discussion 10 Worksheet — **Answer Key**

DS 120 Staff

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Formula Bank

- **Permutations:** $P(n, r) = \frac{n!}{(n-r)!}$
 - **Combinations:** $C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$
 - **Binomial Formula:** $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$
 - **Binomial Probability:** $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
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Warm-Up

1. **Arrange 5 books, selecting 3:**

$$P(5, 3) = \frac{5!}{(5-3)!} = \frac{120}{2} = \boxed{60}$$

2. **Choose 4 from 10 students:**

$$C(10, 4) = \frac{10!}{4!(6!)} = \frac{5040}{24} = \boxed{210}$$

Permutations and Combinations

1. **4 mutual funds from 30, one per quarter:**

$$P(30, 4) = \frac{30!}{26!} = 30 \times 29 \times 28 \times 27 = \boxed{657,720}$$

2. **IRS auditing 3 of 18:**

$$C(18, 3) = \frac{18!}{3!(15!)} = \frac{18 \times 17 \times 16}{6} = \boxed{816}$$

3. Powerball combinations:

$$\text{Choose 5 of 53} \Rightarrow \binom{53}{5} = 2,869,685$$

$$\text{Choose 1 Powerball (1-42)} \Rightarrow 42$$

$$\text{Total possibilities: } 2,869,685 \times 42 = \boxed{120,526,770}$$

4. Senate committee of 5 from 100:

$$\binom{100}{5} = \frac{100!}{5!95!} = 75,287,520$$

If no state has both senators:

$$\text{Choose 5 states: } \binom{50}{5} = 2,118,760, \quad \text{then 1 senator per state: } 2^5 = 32$$

$$\text{Total: } 2,118,760 \times 32 = \boxed{67,800,320}$$

5. Distinct arrangements of PERSEVERE:

Total letters: 9, E appears 4 times, R twice.

$$\text{Arrangements: } \frac{9!}{4!2!} = \frac{362,880}{48} = \boxed{7,560}$$

Binomial Probability Formula

Problem 1 — Horse Racing Favorites

Let $n = 5$, $p = 0.67$.

$$(a) P(X = 2) = \binom{5}{2}(0.67)^2(0.33)^3 = 10(0.4489)(0.0359) \approx \boxed{0.161}$$

$$(b) P(X = 4) = \binom{5}{4}(0.67)^4(0.33)^1 = 5(0.201)(0.33) \approx \boxed{0.331}$$

$$(c) P(X \geq 4) = P(4) + P(5) \quad P(5) = (0.67)^5 = 0.135 \Rightarrow 0.331 + 0.135 = \boxed{0.466}$$

$$(d) P(2 \leq X \leq 4) = P(2) + P(3) + P(4) \quad P(3) = \binom{5}{3}(0.67)^3(0.33)^2 = 10(0.301)(0.109) = 0.328 \\ \Rightarrow 0.161 + 0.328 + 0.331 = \boxed{0.820}$$

Problem 2 — Families with Five Children

Each child has equal chance of being a girl ($p = 0.5$).

$$P(3 \text{ girls, } 2 \text{ boys}) = \binom{5}{3}(0.5)^3(0.5)^2 = 10(0.03125) = \boxed{0.3125}$$

Problem 3 — Baseball Player

(a) Bernoulli trials apply if each at-bat is independent, has two outcomes (hit or not), and the hit probability $p = 0.26$ remains constant.

(b) $P(X \geq 2) = 1 - [P(0) + P(1)]$ $P(0) = (0.74)^4 = 0.300$, $P(1) = \binom{4}{1}(0.26)(0.74)^3 = 4(0.26)(0.405) = 0.421 \Rightarrow 1 - (0.300 + 0.421) = \boxed{0.279}$